

Chapter 6

Damped Oscillation

Damped oscillation is a common phenomenon in physics. We often observe vibrations in which the amplitude gradually decreases, eventually ceasing completely. In real world conditions, in addition to the force that causes the vibration, there is also a force that resists or opposes the vibrational motion. So that, the energy of all vibrating systems eventually decreases and the vibration comes to a stop. In this experiment, you will investigate the damping coefficient of a vibration using two oscillation methods.

6.1 Objectives

1. To determine the damping coefficient of oscillation using two methods, namely: horizontal rod oscillation and spring oscillation.
2. To compare the two methods of measuring the damping coefficient of oscillation.
3. To determine the relationship between the damping coefficient of oscillation and the area of the damping plate used.

6.2 Tools

1. Horizontal rod oscillation system set-up (system equipped with a reflective mirror, light source, horizontal screen, and marker pen).
2. Spring oscillation system set-up (stand, spring, and mass).

3. Aluminum plates with various surface areas.
4. Stopwatch.
5. Measuring tape

6.3 Theoretical Foundation

6.3.1 Horizontal Rod Oscillation Method

As a simple model, we assume a damped oscillation with a damping force proportional to the velocity of the object. The displacement that occurs in this oscillation is in the form of an angular change (θ). The equation of motion for the horizontal rod oscillation can be written as:

$$I \frac{d^2 \theta}{dt^2} + a \frac{d\theta}{dt} + k\theta = 0 \quad (6.1)$$

with I is the moment of inertia of the oscillating system, θ is the angular displacement of the horizontal rod from the equilibrium position, a is the damping constant and k is the elasticity of the system.

The solution to equation (6.1) strongly depends on the values of I , a and k . If the damping is relatively small, the system will still oscillate, although it will eventually come to rest. This condition is referred to as underdamping, and it is the case most emphasized in the damped oscillation experiment. If the damping is increased to the point where the system no longer oscillates, the system reaches critical damping. Similarly, in the case of over damping, the system also does not oscillate. If ϕ is the initial phase of the oscillation, then the solution for the underdamped condition is:

$$\theta = \theta_0 e^{-\gamma t} (\omega t - \phi) \quad (6.2)$$

with

- $\gamma = \frac{a}{2I}$
- $\omega^2 = \omega_0^2 - \gamma^2$
- $\omega_0^2 = \frac{k}{I}$

with γ is the damping coefficient of the oscillation, ω represents the damped oscillation frequency and ω_0 represents the natural frequency of the oscillation.

6.1 Spring Oscillation Method

Equivalent to the horizontal rod oscillation, the system is assumed to undergo damped oscillation with a damping force proportional to the velocity of the object. The difference lies in the fact that the displacement occurs in this oscillation in the form of a change in spring extension (x). Thus, the equation of motion for the spring oscillation is written as:

$$m \frac{d^2x}{dt^2} + a \frac{dx}{dt} + kx = 0 \quad (6.3)$$

with m is the mass of the oscillating load, a is the damping constant, and k is the spring constant. As with the horizontal rod oscillation, the solution to equation (??) strongly depends on the values of m , a , and k . The solution for the underdamped case is:

$$x = x_0 e^{-\gamma t} \cos(\omega t - \phi) \quad (6.4)$$

with

- $\gamma = \frac{a}{2m}$
- $\omega^2 = \omega_0^2 - \gamma^2$

- $\omega_0^2 = \frac{k}{m}$

with γ is the damping coefficient of the oscillation, ω represent the damped oscillation frequency, and ω_0 represents the natural frequency of the oscillation.

From both solutions (6.2 and 6.4), two aspects should be noted, namely the presence of an exponential function factor and a cosine function factor. The exponential function factor determines how quickly the system is damped. The greater the damping ratio, the faster the system decays to zero. In contrast, the cosine function factor indicates that oscillations still occur in the system, with a damped oscillation frequency that differs from the undamped case.

6.4 Experimental Methods

6.4.1 Horizontal rod oscillation

The horizontal rod system set-up consists of a vertical cylinder positioned between two vertical wires. Inside the cylinder, there is a hole for inserting the rod, which is clamped in place. At the end of the horizontal rod, a thin aluminum plate can be attached to serve as a damper. A laser light spot can be projected onto a screen through a mirror mounted on the vertical cylinder. In this way, the oscillation displacement can be measured based on the reflected oscillation on the horizontal screen.

Step of the experiment

1. Set-up the experiment as shown in figure 6.1.

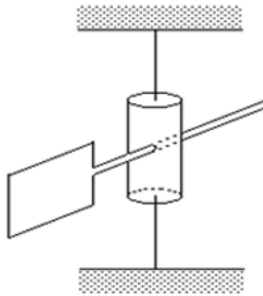


Figure 6.1: Set-up of horizontal rod oscillation

2. Prepare a stopwatch to measure the time required for 20 oscillations.
3. Displace the rod by a small angle. Be ready to start the stopwatch as the rod is released.
4. Mark the horizontal screen (using a marker) at each displacement made by the oscillating rod on one side of the swing.
5. Measure the displacement and record them in the lab table.
6. The marks on the horizontal screen can be cleaned and repeat steps 1-5 using aluminum plates of different surface areas.
7. From the collected data set, plot a graph of displacement as a function of time.
8. Use the appropriate graph equation so that the damping coefficient of the oscillation (γ) can be determined from the graph. Perform this calculation for each plate surface area used.
9. Create a graph of the damping coefficient (γ) as a function of the plate surface area. Use the correct graph of the damping to show how the damping coefficient

relates to the surface area of the damping plate using the rod oscillation method.

6.4.2 Spring Oscillation Method

In the second system, the oscillator set-up consists of a load mass suspended from a spring, and a horizontal plate attached to the load. This plate surface functions as the damper.

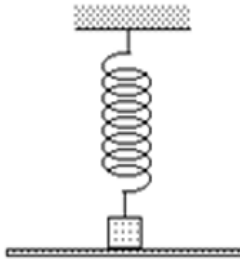


Figure 6.2: Set-up of spring oscillation

Step of the experiment

1. Set-up the experiment as shown in figure 6.2.
2. Prepare a stopwatch and a ruler to measure the spring displacement.
3. Displace the load mass from its equilibrium position and release it so that the spring mass system with the attached plate oscillates.
4. Record the displacement for each oscillation of the spring. If direct measurement of displacement is difficult to perform, use a white paper as a vertical screen placed installed beside the spring and mark (using a pen) for each displacement that occurs.
5. Also record the total time for 20 oscillations.

6. Repeat steps 1-5 for different plate surface areas.
7. As in the horizontal rod oscillation method, plot the displacement as a function of time. Use the correct graph equation so that the damping coefficient (γ) can be calculated from the graph. Perform this calculation for each plate area used.
8. Create a graph of the damping coefficient (γ) as a function of the plate surface area. Use the correct graph equation to show the relationship between the damping coefficient and the surface area of the damping plate using the spring oscillation method.
9. As the final step, compare the damping coefficient measurements using the horizontal rod oscillation method with the spring oscillation method.